

Nicholas Rios

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EDUCATION

University of Texas at Arlington, College of Engineering - Arlington, TX

Expected Dec 2026

Bachelor of Science in Mechanical Engineering | GPA: 3.5

ENGINEERING EXPERIENCE

Lockheed Martin Missiles and Fire Control | CWEP Mechanical Engineer

July 2025 - Present

- Support multidisciplinary aerospace programs from concept through prototype fabrication, resolving design and manufacturability challenges for functional hardware and laboratory improvement projects.
- Create and revise Creo CAD models for fixtures and commissioned prototypes, incorporating design intent, assembly interfaces, fabrication methods, and design-for-manufacturing requirements.
- Manufacture and inspect prototype components using CNC routers, waterjet and laser cutting systems, plus FDM, composite, and SLA additive processes; perform fit checks and implement design revisions.
- Coordinate concurrent requests with engineers across multiple programs and communicate design and fabrication status while following safety, quality, and information-security requirements.

ENGINEERING PROJECTS

3D-Printed Autorotating Drone | Materials and Test Lead

Spring/Summer 2026

- Won the 2026 3DPAC Rotary-Wing Competition with a semi-autonomous UAV featuring a fully 3D-printed airframe, powered ascent, and three-blade autorotation recovery.
- Designed and iterated rotor blades, hub components, and airframe assemblies in Onshape, evaluating structural load paths, assembly fit, manufacturability, material constraints, and system interfaces.
- Formulated a nonlinear rotor-geometry optimization using AeroSandbox and CasADi/IPOPT, increasing average hang time 240% relative to the baseline design.
- Led material selection and validation for ASA Aero, ASA, and Onyx components through fit checks, dry assembly, ground spin tests, structural checks, and flight testing; documented results to guide design revisions.

Formula SAE Radiator Optimization | Simulation Lead

Spring 2026

- Developed and documented effectiveness-NTU calculations and ANSYS Fluent thermal-fluid simulations to evaluate heat rejection, airflow, and packaging under expected vehicle operating conditions.
- Reviewed simulation results and demonstrated the feasibility of consolidating a dual-radiator system into one larger radiator while meeting thermal and packaging requirements.

Cycloidal Drive Structural Analysis | Mechanical Design Project

Spring 2025

- Built static and transient ANSYS Mechanical models for cycloidal-drive components, defining contacts, supports, and operating loads and reviewing stress and deformation results to assess design performance.
- Compared solution approaches and refined contact, support, and mesh definitions to obtain stable, physically credible stress and deformation results.

Ball-Balancing Platform | Mechanical Lead

Fall 2025

- Designed the mechanical structure and actuation platform responsible for dynamically tilting a plate to control position of a ball.
- Collaborated with controls and electronics team members to integrate sensors, actuators, and microcontroller control logic using a Raspberry Pi Pico.

TECHNICAL SKILLS

CAD and Drawings: SolidWorks, Creo, Onshape; ASME Y14.5 GD&T; manufacturing drawings; DFM

Analysis: ANSYS Mechanical and Fluent (FEA/CFD); MATLAB/Simulink; heat-transfer calculations; traditional stress analysis

Manufacturing and Test: FDM, SLA, CNC routing, waterjet and laser cutting; fit checks and prototype testing